

Question 1:

I agree <Tên methodology> methodology, here's why:

a) Requirements characteristics:

- Reliability: The project has clear and achievable requirements. It will be upon completion and anticipated to deliver a high level of reliability
- Type and number of requirements:
 - + This project has two types of requirements: 5 Functional Requirements, and 5 Non - Functional Requirements.
 - + Since the projects are well-defined but not enough, they may evolve and be clarified further during the project. Therefore, the project should be Scrum Development methodology - which supports iterative development and flexibility to accommodate changing and incomplete requirements.
- Determination of Requirements at an Early Stage:
 - + <Tên hệ thống phát triển> is a newly developed system, and this is the development team's first time working on the project. Therefore, the time requires additional personnel to participate and supports the project to ensure high-quality completion.

b) Development team:

- Team size: The development team includes 3-4 extensive experienced and skilled developers. This is a <Mức độ> project.
- Level of understanding of user requirements: The development team has extensive experience and skills, so they can easily understand the requirements in this project.

c) User involvement:

- Our team is supported by <Tên stakeholder chính>. Other departments have committed to sending <số lượng/người/bộ phận hỗ trợ> to join and support our team. This ensures the organization to provide additional resources and information when necessary, and user involvement in this project is high.

Conclusion:

- I agree using <Tên methodology> methodology is suitable for this project. As this is a <Mức độ>-level project, the development team should break it down into smaller, manageable parts to simplify the process. Additionally, since this is a newly developed system and user requirements may evolve, the project should remain flexible to adapt to changes. Our team is supported by <Tên stakeholder chính>, and other departments have committed to assigning <số lượng/người/bộ phận hỗ trợ> to join and support our team throughout the project.

I disagree <Tên methodology> methodology, I recommend using Agile/Scrum. Here's why:

d) Requirements characteristics:

- Reliability: The project has clear and achievable requirements. It will be upon completion and anticipated to deliver a high level of reliability
- Type and number of requirements:
 - + This project has two types of requirements: 5 Functional Requirements, and 5 Non - Functional Requirements.
 - + Since the projects are well-defined but not enough, they may evolve and be clarified further during the project. Therefore, the project should be Scrum Development methodology - which supports iterative development and flexibility to accommodate changing and incomplete requirements.
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 - + <Tên hệ thống phát triển> is a newly developed system, and this is the development team's first time working on the project. Therefore, the time requires additional personnel to participate and supports the project to ensure high-quality completion.

e) Development team:

- Team size: The development team includes 3-4 extensive experienced and skilled developers. This is a <Mức độ> project.
- Level of understanding of user requirements: The development team has extensive experience and skills, so they can easily understand the requirements in this project.

f) User involvement:

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Conclusion:

- I recommend using <Tên methodology> methodology is suitable for this project. As this is a <Mức độ>-level project, the development team should break it down into smaller, manageable parts to simplify the process. Additionally, since this is a newly developed system and user requirements may evolve, the project should remain flexible to adapt to changes. Our team is supported by <Tên stakeholder chính>, and other departments have committed to assigning <số lượng/người/bộ phận hỗ trợ> to join and support our team throughout the project.

Question 2:

1. Type and level/ stage of testing to meet customer's requirements:

- Unit Testing - to detect bugs early in individual components.
- Integration Testing - to ensure modules interact and exchange data properly.
- System Testing - to validate the entire system works correctly.
- Acceptance Testing - to confirm the system meets customer/ business requirements.

2. Who performs testing at each level/ stage?

- Unit Testing: Developers - they know the internal logic and can verify details at the smallest level.
- Integration Testing: Developers/ Testers - developers check technical integration, tester validate end-to-end interaction.
- System Testing: QA/ Testers - independent testers verify the full system against requirements.
- Acceptance Testing: End-users/ Clients - only customers can confirm whether the system meets business needs.

Question 5:

1. Layered Architecture:

- The system is divided into layers (e.g., UI, Business Logic, Data).
- Each layer only interacts with the adjacent one.
- Easy to maintain, modify, and reuse.

2. Client-Server Architecture:

- Consists of Client (sending requests) and Server (processing and responding).
- Centralized data management, supports multiple clients, and is scalable.

SP25

5 Quality Metrics for Code and Design:

1. Cohesion (LCOM): Measures how closely related methods in a class are. Lower LCOM → higher cohesion (better).
2. Coupling (CBO): Measures how many classes a class depends on. Lower coupling → better modularity and easier maintenance.
3. Cyclomatic Complexity: Counts independent paths in a program. Lower values → simpler and more understandable code.
4. Code Coverage: Percentage of code executed during testing. Higher coverage → better-tested code.
5. Lines of Code (LOC): Counts lines in the codebase. Helps estimate complexity and maintenance effort.

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- a) Velocity: Number of story points completed in one iteration.
- b) Story Point: Unit for measuring task complexity/effort.
- c) Example: Total story points = 120, Velocity = 20 $\rightarrow$ Iterations = $120 \div 20 = 6$ iterations.

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Predictive Software Development Models:

- Waterfall Model: Linear, sequential phases (requirements $\rightarrow$ deployment).
- V-Model: Extension of Waterfall, each dev phase has a matching test phase.
- Spiral Model: Iterative cycles (planning, risk analysis, engineering, evaluation) with focus on risk management.

Adaptive Software Development Models:

- Agile Model: Iterative, customer collaboration, adapts to changing requirements.
- Scrum: Agile framework with sprints, defined roles (Scrum Master, Product Owner), and regular ceremonies.
- Extreme Programming (XP): Agile practice-focused (CI, TDD, frequent releases) to improve quality and responsiveness

As a <user>, I want <something>, so that I can <benefit>

Question 6:

A. Activity and User task

1. Make an appointment
 - 1.1 select services/ packages
 - 1.2 choose date and time
 - 1.3 choose employees
 - 1.4 view estimated amount
 - 1.5 receive successful booking notification
2. Manage Payment
 - 2.1 choose method for payment
 - 2.2 process payment

B. Release

Release 1:

- 1.1.1 As a Customer, I want select services, so that I can select best service I want
- 1.2.1
- 1.3.1

1.4.1

Release 2:

1.5.1 As a customer, I want to receive successful booking notification, so I can know whether an appointment can be made or not

2.1.1

2.2.1